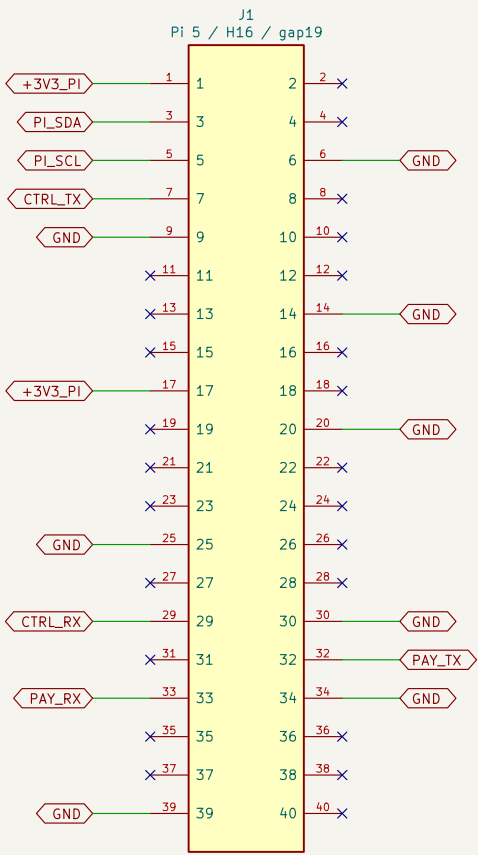
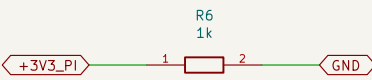


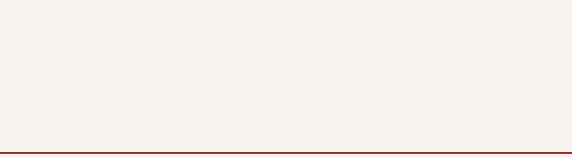
RBPHAT / REV B4 XA Dual full-duplex UART + 3S battery monitor



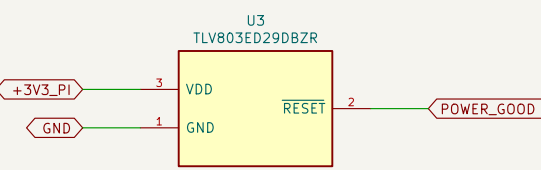
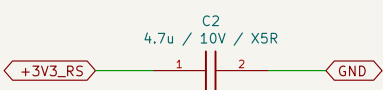
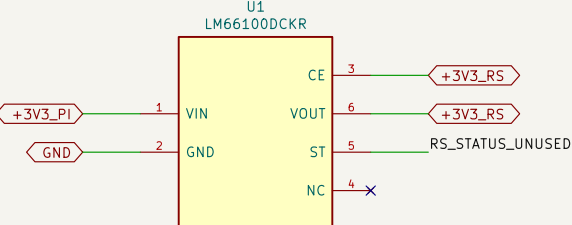
5 V and ID EEPROM pins unused.
UART2: GPIO4/5; UART4: GPIO12/13.
Header body 16 mm; PCB gap 19 mm; Active Cooler below.



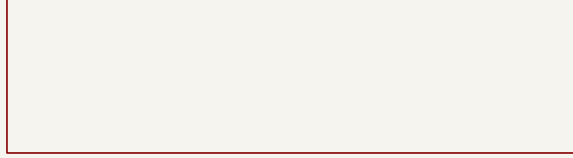
CTRL full-duplex RS-422



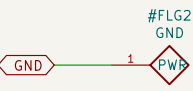
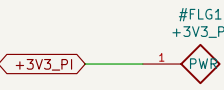
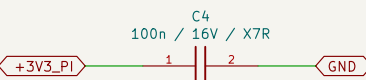
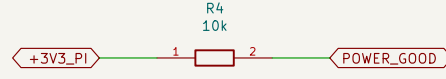
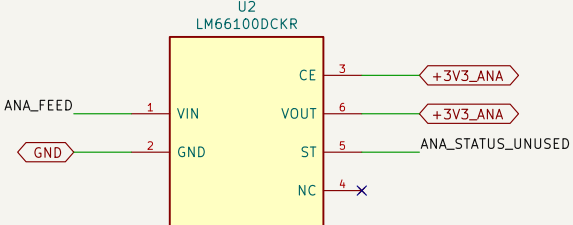
File: control.kicad_sch



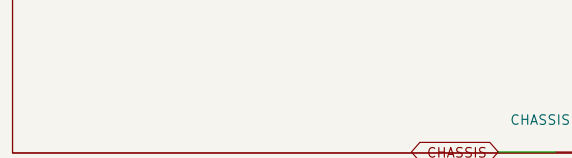
PAY full-duplex RS-422



File: payload.kicad_sch



Battery sensing and I2C isolation



File: battery.kicad_sch

LM66100 CE = VOUT: reverse blocking. POWER_GOOD: 2.93 V supervisor, 50 ms nominal release.
C3 maintains ADC supply during switch opening. Validate fast power collapse on prototype; see DESIGN_REV_B.md.

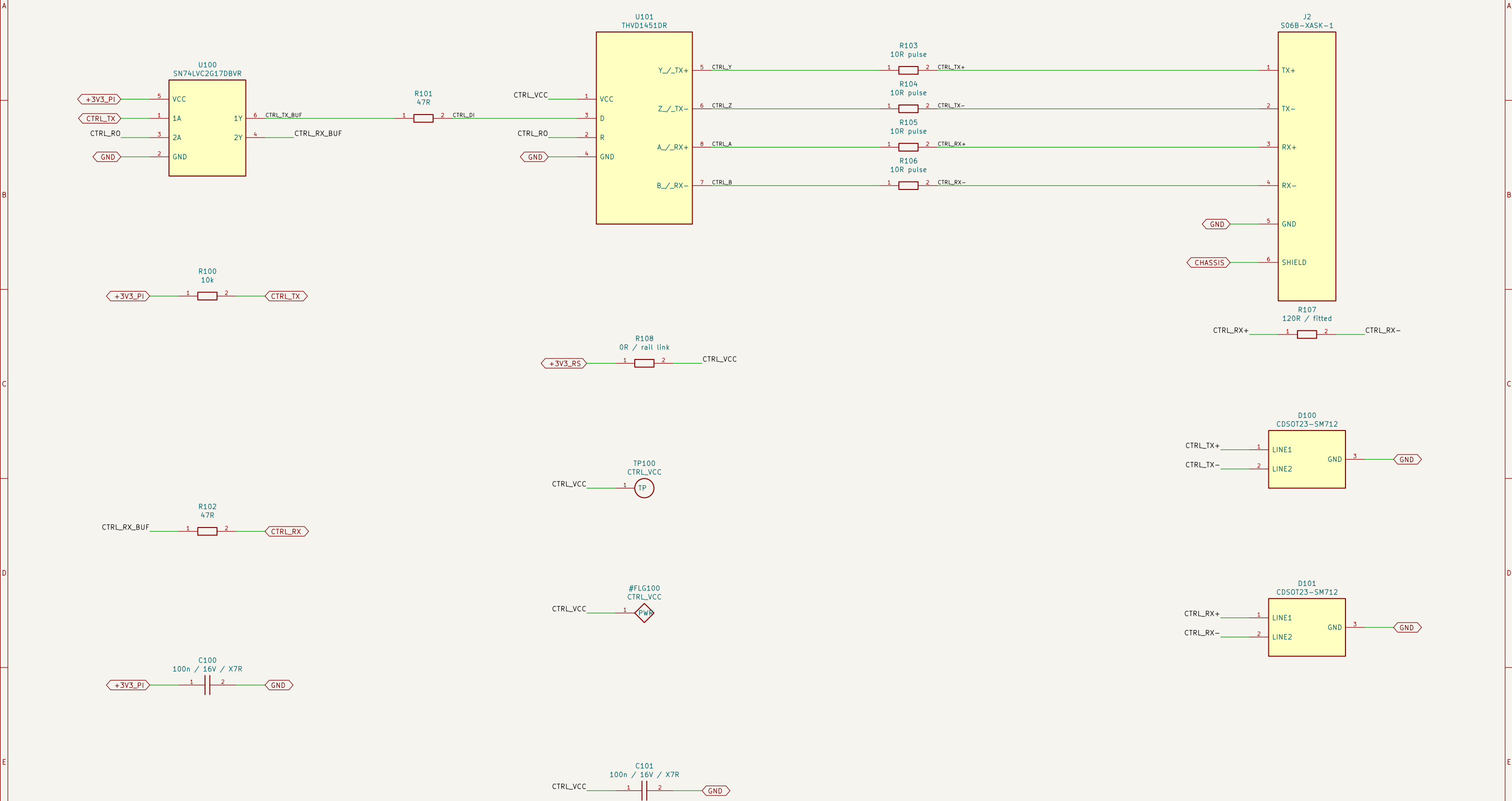
H16 socket / gap19 / Active Cooler; common ground; prototype / B4 XA horizontal
LOONAR

Sheet: /
File: rbphat.kicad_sch

Title: Pi 5 interface and protected power

Size: A3	Date: 2026-09-11	Rev: B4
KiCad E.D.A. 10.0.4		Id: 1/4

CTRL / fail-safe receiver / no external bias / 2 Mbit/s target



TX pair terminates at remote receiver. Local 120R fitted. Pin 6 SHIELD -> isolated CHASSIS lug SH1.
Connector -> TVS -> 10R -> transceiver. Reference: TI THVD14xx Figure 38 / Table 7.
Common ground; cross TX to remote RX. No galvanic isolation. Protection rating requires board-level testing.

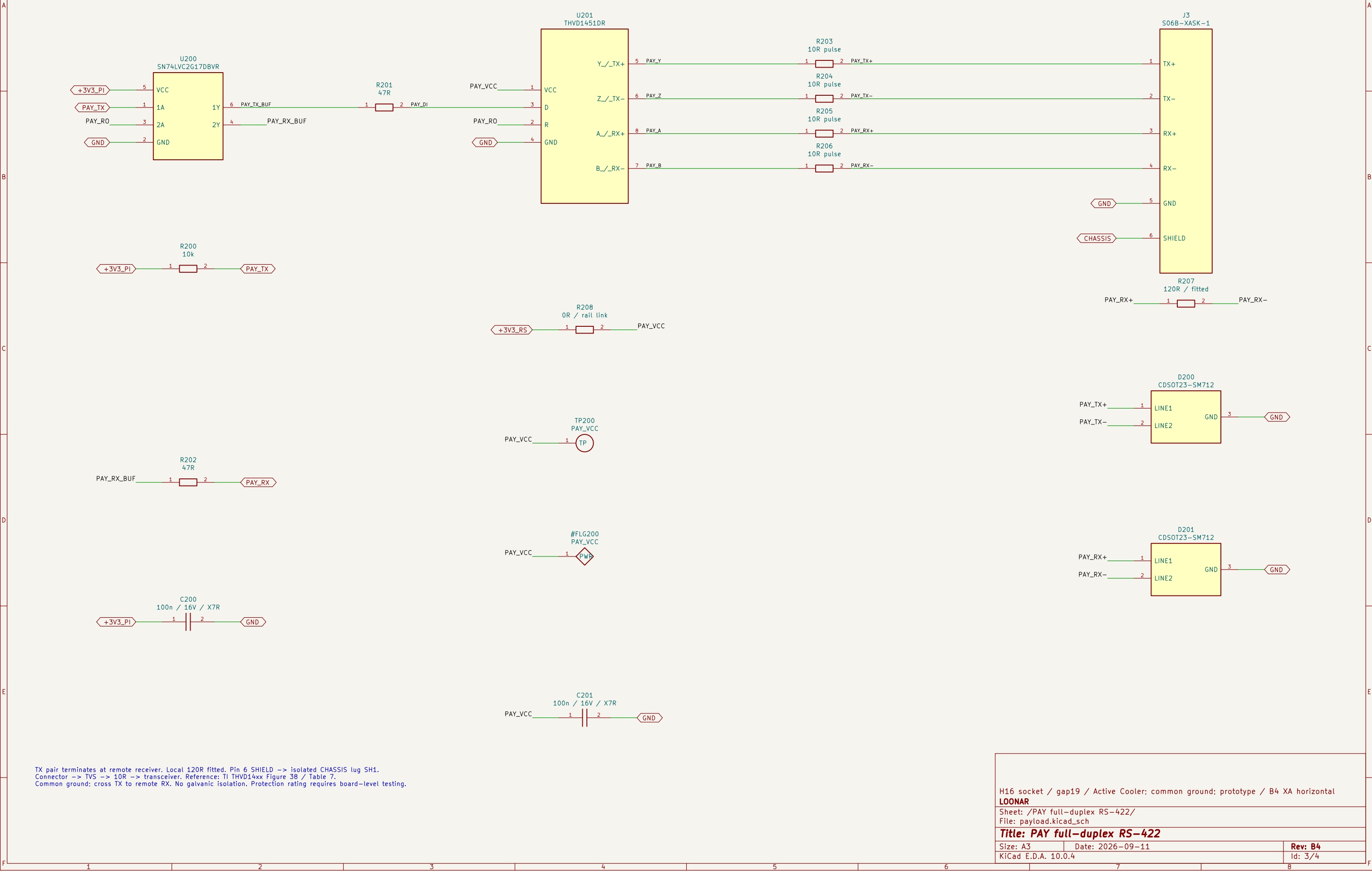
H16 socket / gap19 / Active Cooler; common ground; prototype / B4 XA horizontal
LOONAR

Sheet: /CTRL full-duplex RS-422/
File: control.kicad_sch

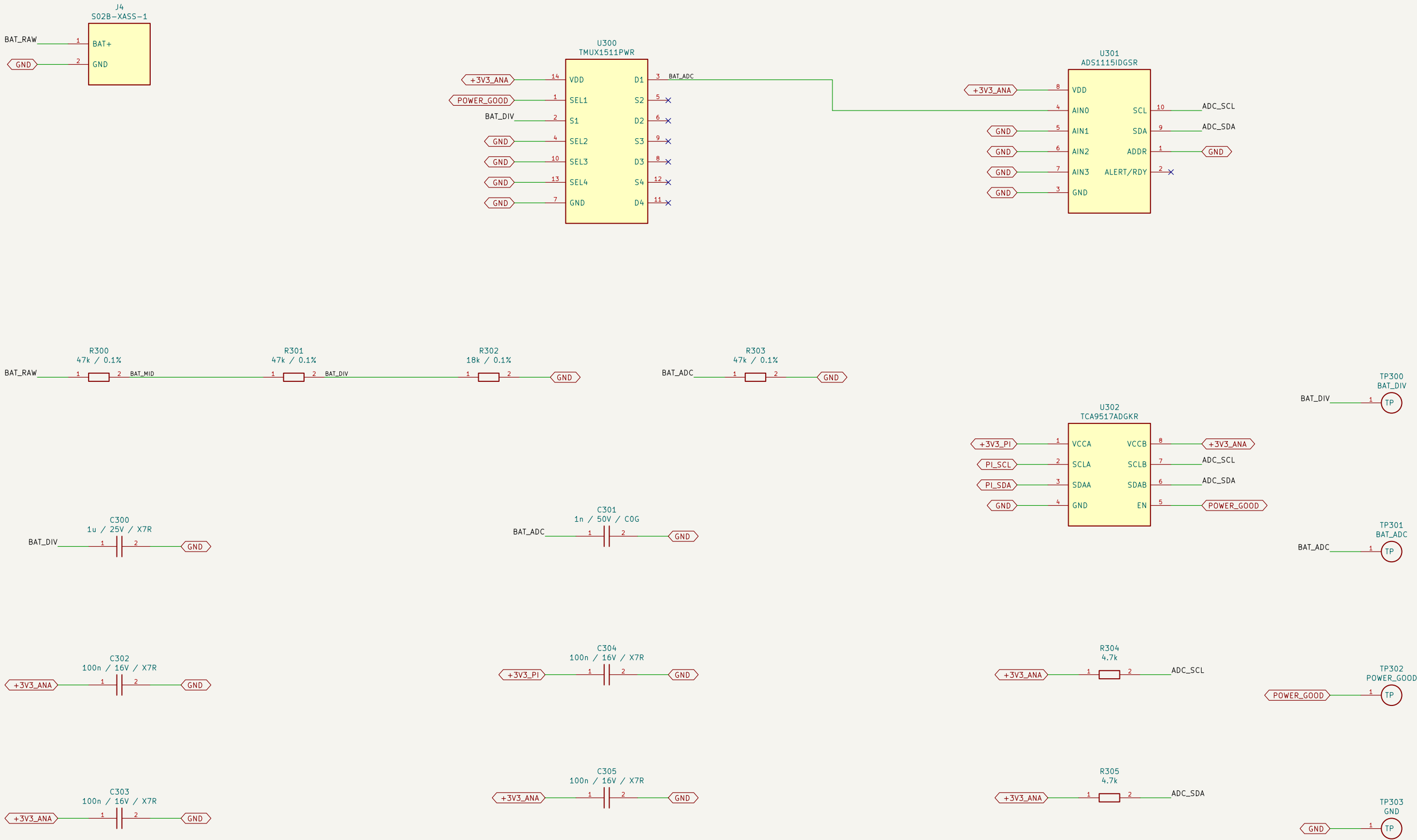
Title: CTRL full-duplex RS-422

Size: A3	Date: 2026-09-11	Rev: B4
KiCad E.D.A. 10.0.4		Id: 2/4

PAY / fail-safe receiver / no external bias / 2 Mbit/s target



3S battery / 0 ... 12.6 V / ADS1115 address 0x48 / PGA +/-4.096 V



Nominal VBAT = ADC_V * 8.222222 (includes R303: Ron and ADC loading not included).
ADC off leakage discharges through R303. POWER_GOOD enables switch and I2C together.
Wait >= 150 ms after enable before measurement. Sense ground must not carry motor return current.

H16 socket / gap19 / Active Cooler; common ground; prototype / B4 XA horizontal
LOONAR

Sheet: /Battery sensing and I2C isolation/
File: battery.kicad_sch

Title: Battery sensing and I2C isolation

Size: A3	Date: 2026-09-11	Rev: B4
KiCad E.D.A. 10.0.4		Id: 4/4